The Impact of Kratom Drinks on Kratom Use: A Trend Analysis

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Abstract

Objectives: To analyze recent trends in the use of kratom drinks and assess the potential public health implications.
Study Design: A retrospective observational study using content analysis of Reddit posts related to kratom.
Methods: Data were collected from the top 20 kratom-related subreddits using the PRAW package in Python. Posts and comments containing keywords related to kratom drinks were extracted and aggregated monthly. The number of posts was normalized by the number of Reddit users to account for user growth over time. Trends were analyzed to identify significant shifts in interest and discussions about kratom drinks.
Results: The analysis revealed an increase in interest in kratom drinks starting in mid-2022, coinciding with their widespread availability in vape shops, bars, convenience stores, and online. These trends indicate growing public health concerns related to the use of kratom drinks.
Conclusions: The study highlights the need for enhanced public awareness of their potential risks.

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1 Introduction

Kratom, scientifically known as Mitragyna speciosa, is a tropical tree from Southeast Asia and is traditionally consumed by chewing, smoking, or brewing as tea (Charoenratana et al., 2021). Modern consumption of kratom is primarily through oral ingestion, using tablets, capsules, or extracts. It has opioid-like properties, producing stimulant effects at low doses and sedative effects at high doses. In the United States, it has historically been used by people with substance use disorder to manage symptoms associated with drug withdrawal (especially opioid, heroin, and fentanyl), pain, fatigue, and mental health issues (Palamar, 2021; Tobacyk et al., 2022; Williams and Nikitin, 2020; Prozialeck et al., 2019).

Although kratom is unscheduled at the federal level and legal in most U.S. states, the U.S. Drug Enforcement Agency (DEA) has identified it as a drug of concern, and the U.S. Food and Drug Administration (FDA) has issued multiple warnings against its use due to ongoing safety evaluations (Palamar, 2021). Despite these warnings, new kratom drinks have been introduced in convenience stores like 7-Eleven and bars.[1] To better understand if these new kratom drinks have created more interest in kratom over time, we examined the trends in Reddit keyword searches of post titles and comments on the most popular 20 kratom-related subreddits.

Increasing trends in kratom use are concerning because of kratom's addictive qualities. Major adverse effects from kratom use have been reported, including agitation, seizures, central nervous system depression, and neonatal abstinence syndrome (Palamar, 2021). Furthermore, kratom abuse has been associated with withdrawal symptoms such as nausea, itching, sweating, dry mouth, constipation, increased urination, tachycardia, vomiting, drowsiness, and loss of appetite (FDA, 2024; Rogers et al., 2024). Many consumers are unaware that these new kratom drinks are potentially addictive and opioid-like. For example, a U.S. District Judge denied 7-Eleven's motion to dismiss claims that it violated California's unfair competition law by selling a tonic drink, Feel Free, containing kratom, a potentially addictive substance. The lawsuit alleges that 7-Eleven engaged in unfair practices by selling the drink without disclosing its harmful ingredients (Courthouse News Service , 2023). Similarly, families of individuals who died after consuming kratom drinks have sued kratom manufacturers, alleging that the companies failed to provide adequate warnings about the risks associated with kratom use (NPR, 2019). Over 580 people in Florida alone have died from kratom-related overdoses in the past decade,

[1] New kratom drinks include: Feel Free, New Brew Euphoric Seltzer, Super Speciosa Super Feels Chill Vibes Kava & Kratom Shot, Koi Kratom Liquid Kratom Shots, Kingdom Kratom Xtreme Kratom Extract Shot, K Tropix Energy Kratom Shot, and Vivazen Liquid Kratom, among others (Seattle Met, 2023).

with 46 deaths attributed solely to kratom (Tampa Bay Times, 2023). These lawsuits and deaths raise concerns about a *potential* public health crisis, similar to the one caused by OxyContin, where many people consumed the drug without fully understanding its addictive nature (American Bar Association, 2019).

While there is a growing literature on kratom use, it remains an understudied area because of data limitations, and we are unaware of any paper that examines the impact of kratom drinks specifically on trends in kratom use. Given the addictive characteristics of kratom, coupled with the increasing interest in kratom as a result of new kratom drinks and the lack of consumer awareness about the risks associated with kratom-related products, it is important that we better understand the trends in kratom use to avoid a *potential* public health concern.

2 Data Collection and Analysis

Data on kratom use is limited because it is not monitored by the FDA, can be purchased over the counter, is typically not prescribed by doctors, and is not commonly included in most public health surveys that rely on self-reported data. Most companies selling kratom are not publicly traded, resulting in limited sales data. The National Survey on Drug Use and Health (NSDUH) began including questions about kratom use in 2019. However, conducting a time series analysis using NSDUH data is challenging due to issues with data pooling related to the introduction of web interviewing. Moreover, most other nationally representative datasets, such as the Medical Expenditure Panel Survey (MEPS), the Behavioral Risk Factor Surveillance System (BRFSS), and the National Substance Use and Mental Health Services Survey (N-SUMHSS), do not include questions about kratom use.

To overcome these data limitations, we utilized the social media platform Reddit. Reddit is an ideal datasource for discussions about substance use because it provides users with an anonymous platform to post their questions or discuss their experiences with addictive substances. Reddit data has been used in various research studies across a wide range of topics, such as COVID-19 and mental health (Zhu et al., 2023), mental health expressions on student loan debt (Sinha et al., 2023), suicide (Joseph et al., 2023), addiction (Kramer et al., 2024), stock market sentiment (Long et al., 2023), and most similar to this paper, kratom use (Smith et al., 2021).

Reddit was founded in 2005. As of 2024, Reddit boasts over 267.5 million weekly active users, making it one of the largest online communities. Each subreddit (community) is

dedicated to a specific topic, ranging from general interests to niche subjects (DIGIDAY, 2024). We obtained subreddit posts from 2017 through the first quarter of 2024. Before gathering the data, we signed up for Reddit credentials to access the API. Using these credentials, we captured information about the top 20 kratom-related subreddits, as shown in Table A1 in the appendix. Utilizing the PRAW package in Python and cross-referencing with the RedditExtractor package in R, we identified 144 subreddits related to kratom. These were then sorted by the number of subscribers. Some subreddits, such as the news subreddit, were not relevant to kratom despite being returned by PRAW. To refine our selection, we manually reviewed the subreddits by searching Reddit for kratom-related communities and chose the most relevant ones with the most followers.

Once the subreddits were selected, we used the PRAW package to search for terms related to kratom drinks and opioid use within these subreddits.[2] The package connected to the Reddit API to retrieve posts mentioning the key terms in either titles or comments, while managing pagination and rate limits. Figure 1 shows the amount of posts returned based on these two categories of search terms.

For top posts, we employed the requests library and the fetch top posts command to specify timeframes and subreddits, focusing on the top 20 subreddits over the past month, year, and all time. It is important to note that top posts vary by timeframe and subreddit. For example, the top post scores on the kratom subreddit from 2017-2024 ranged from 967 to 161, whereas on the KratomKorner subreddit, they ranged from 265 to 14. We implemented a minimum score of 5 and dropped duplicate posts to emphasize the quantity of posts over their popularity, as the latter is influenced by the number of followers rather than reflecting the average user experience.

To examine trends in kratom use over time, we analyzed the top posts about kratom on the top 20 subreddits about kratom use. Top posts, which are determined by their net score (total upvotes minus total downvotes), indicate the popularity and perceived value of a post. We categorized each subreddit into four distinct groups: General Discussion, Complements/Substitutes, Growing/Buying, and Kratom Addiction. This classification helps to identify the types of posts being shared.

After collecting information on the top posts, we conducted a word analysis on posts categorized under General Discussion. A machine learning workflow using Python and scikit-learn classified the post titles into nine categories listed in Table 1. We manually created a legend of 1,000 post categories to guide the machine-learning process, using a

[2]For more information about the PRAW package, please see: https://praw.readthedocs.io/en/stable.

confidence threshold of 0.6. Post-categorization was manually verified to ensure accuracy and uncategorized posts were reviewed for potential inclusion. All posts were analyzed. These classifications were then manually checked by the researchers in this study. Data on the monthly active users is an estimation and based on several data sources (Curry, 2024; Dean, 2024).[3]

2.1 Limitations

The most significant limitation of this study is the lack of detailed information about kratom-infused beverages. If users are unaware that these drinks contain kratom, they are unlikely to post about them, leading to underreporting. This limitation is inherent in any dataset relying on self-reported data. Additionally, Reddit is not a representative sample of the general population, which introduces potential self-selection bias in our trend analysis. We also lacked access to demographic data, such as race, ethnicity, or gender, due to the nature of Reddit data. Another limitation is the absence of an analysis of causal mechanisms. While we can observe general trends in kratom use, we cannot conclude that these new kratom beverages are causing public health concerns. Addressing this gap presents an important avenue for future research. Similar to Smith et al. (2021) this study should be a starting point for researchers to fully understand the impact of kratom drinks on kratom use.

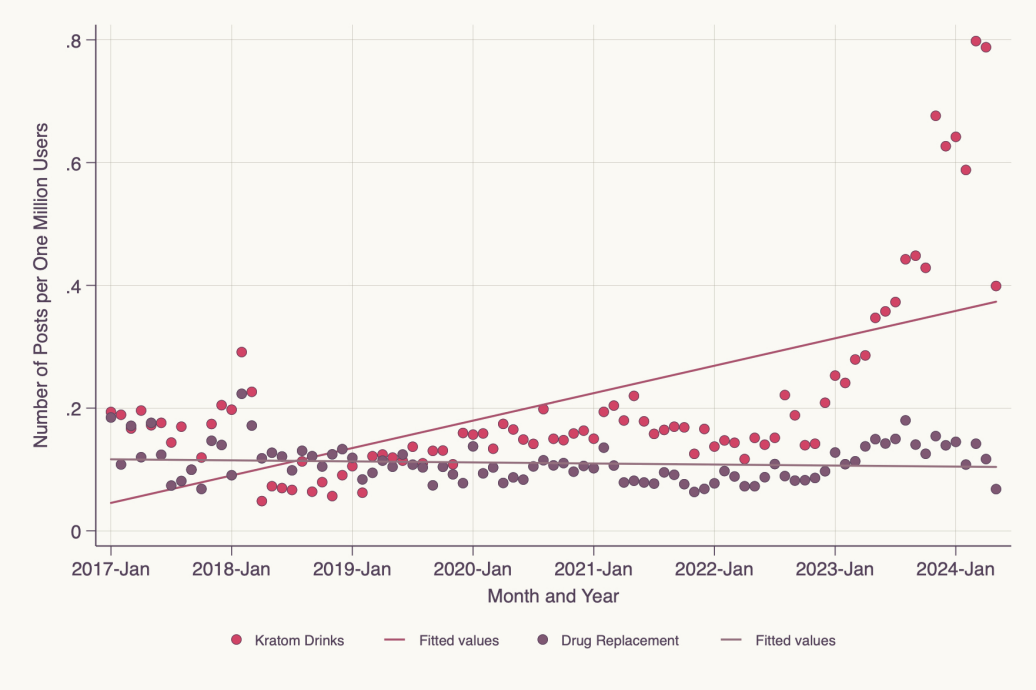
3 Study Results

Figure 1 illustrates the trends in searches for kratom drinks compared to searches for heroin, fentanyl, OxyContin, painkillers, and other related substances on kratom subreddits from 2017 to 2024.

To account for the increasing number of Reddit users, we normalized the data by dividing the number of posts by the number of millions of users.[4] The trends in searches for kratom drinks across kratom subreddits remained relatively stable until mid-2022, when there was a sharp increase in the number of posts per million users. This upward trend has continued into 2024. As a comparison, the search trends per million users for opioid like substances on kratom subreddits remained stagnant, with a slight downward trend over time. These new drinks, targeting individuals who are sober but seek a "feel-good" effect,

[3]While the estimates of monthly active users vary across data sources, the trends remain linear and demonstrate significant user growth over the study period, thus not affecting the study results.

[4]This normalization ensures that observed trends reflect genuine shifts in interest or behavior rather than just fluctuations in user numbers. It allows for a fair comparison of post activity across different time periods, accurately assessing the relative popularity of kratom-related discussions.



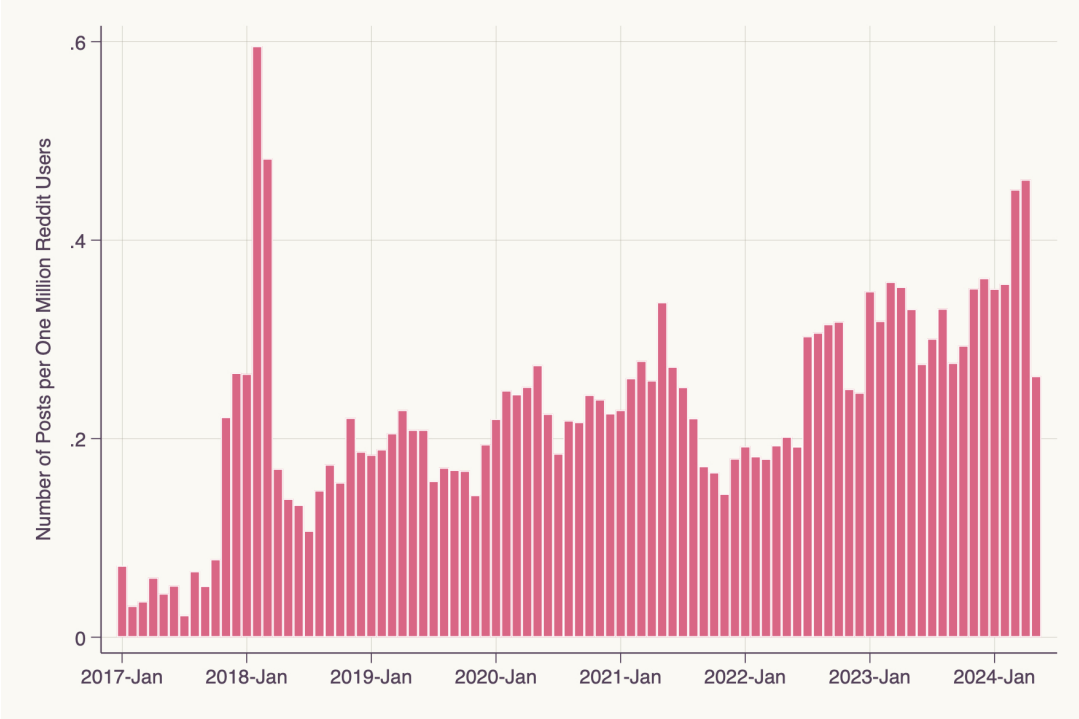
Notes: Data is normalized by dividing the number of posts by the number of millions of users.

Figure 1. Trends in Reddit Posts by Keyword Search

include tea, smoothies, and shots (Seattle Met, 2023). Its use in the U.S. has increased, with widespread availability in vape shops, bars, convenience stores, and the Internet (Rogers et al., 2021; Bowdring et al., 2023). For example, around 46.2 percent of vape shops in California offer kratom products (Bowdring et al., 2023). Figure 1 highlights that the increased availability of kratom drinks has led to increased interest in kratom.

Figure 2 shows the overall trends in interest in kratom over time by examining the number of top posts per million monthly users by month and year. There has been an increasing trend in the number of top posts per million monthly users, with a notable surge in interest at the end of 2017 and the beginning of 2018. This spike coincides with the crackdown on opioid prescriptions, prompted by the record-high opioid-related deaths in 2017, and the DEA's proposal to classify kratom as a Schedule I drug in late 2016 (NIH, 2024; Castillo, 2018).

To understand the content of these posts, Figure 3 illustrates the variation in the total number of top posts across different time periods and four subreddit categories. In the complements/substitutes subreddits, discussions often revolve around other natural supplements like kava, khat, and kanna, which are commonly used alongside kratom. Posts in the growing/buying kratom subreddits typically focus on where to purchase kratom and how to cultivate it, frequently highlighting specific companies and products. Subreddits dedicated to kratom addiction offer support for individuals trying to quit kratom



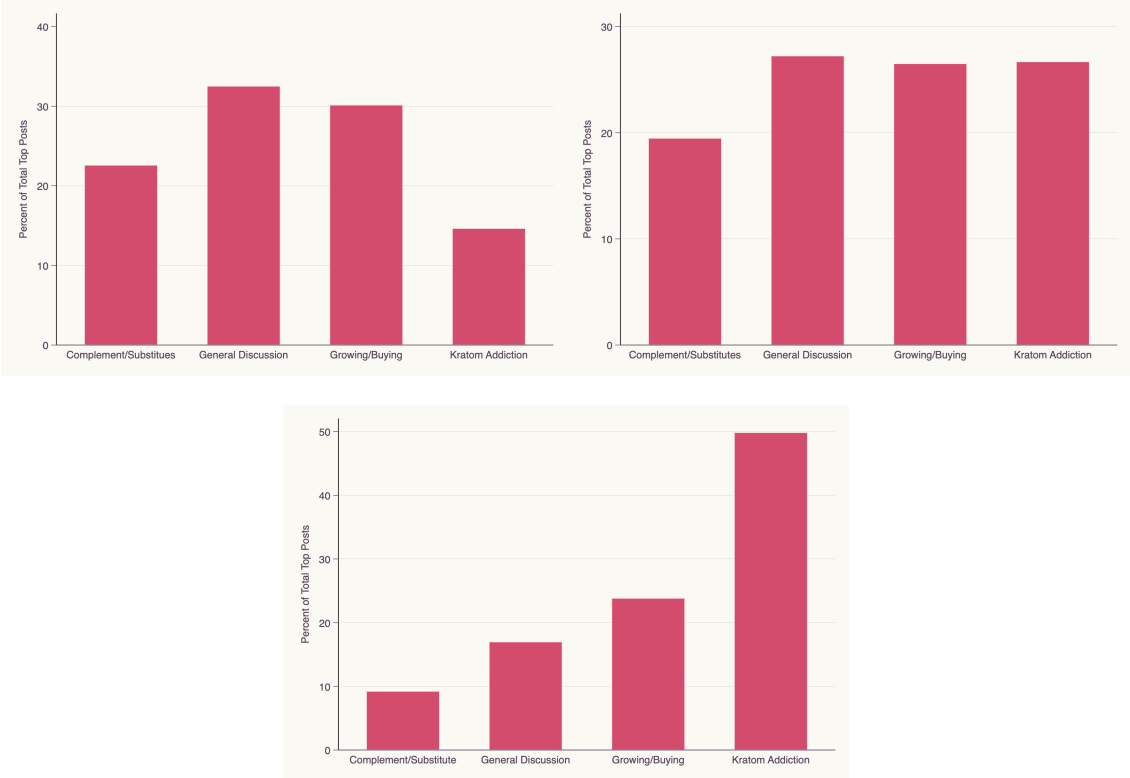
Notes: Data is normalized by dividing the number of posts by the number of millions of users. We only kept top posts with a score above four and we deleted duplicate posts from the sample.

Figure 2. Number of Top Posts in Kratom Related Subreddits

and manage withdrawal and tolerance. General discussion subreddits encompass a wide range of topics, including those mentioned above and more, catering to various interests related to kratom.

From 2017 to 2024, the majority of top posts were concentrated in subreddits dedicated to general discussions about kratom. Subreddits focusing on growing or purchasing kratom comprised the second-largest proportion of posts across all three time periods. Although posts about kratom addiction represented the smallest percentage of kratom-related subreddit activity from 2017 to 2024, this percentage has gradually increased such that in our most recent time period posts on subreddits about kratom addiction made up the largest proportion of top posts.

To gain a deeper understanding of the content in general discussion kratom subreddits, we analyzed the post titles and categorized them into nine distinct groups derived from (Smith et al., 2021), and shown in Table 1. Table 1 reveals that discussions on kratom regulation constitute the largest portion of top posts over the past year and the previous seven years. Although kratom regulation has been tabled by the DEA and FDA, numerous states, cities, and counties have attempted to pass bills banning kratom. Most of these posts urge Reddit users to sign petitions, contact local representatives, and share their personal experiences with kratom use.



Notes: The first panel shows the percent of posts by subreddit category from January 2017 to May 2024. The second panel displays the percent of posts by subreddit category from May 2023 to May 2024. The final panel represents the percent of posts by subreddit category from April 21, 2024 to May 20, 2024. Data is normalized by dividing the number of posts by the number of millions of users.

Figure 3. Changes in Trends of Top Posts on Kratom Subreddits by Type of Subreddit

The second most common type of post discusses using kratom as a substitute for other drugs or for pain management. Many posts inquire about using kratom to alleviate withdrawal symptoms from opioids or alcohol, maintain abstinence from other substances, and manage chronic or temporary pain. Kratom addiction is the third most frequent discussion topic among top posts over the past seven years and the fourth most common in the past year's top posts. Given the subreddits dedicated to kratom addiction, these posts often examine what kratom addiction looks like, how to identify if a user is addicted, and how to cope with tolerance build-up or withdrawal symptoms. The trend shifts in the most recent month's top posts, with adverse health effects becoming the main topic, followed by polydrug use. Common complaints about kratom include kratom wobbles, nausea, constipation, impacts on sex, and effects on sleep. Posts about polydrug use frequently seek others' experiences mixing kratom with other substances to ensure their safety.

Over time, the topics discussed in kratom-related posts have evolved. Posts about adverse health effects as a proportion of total posts have increased the most, followed by discussions on polydrug use, kratom dosing, and reporting psychiatric symptoms or his-

Table 1. Top Posts in General Discussion Kratom Subreddits

	Jan 2017 – May2024	May 2023 – May 2024	April 2024 – May 2024
Total Posts for Each Topic			
Adverse Health Effects	92	63	18
Desirable Health Effects	71	17	2
Kratom Addiction	133	51	8
Kratom as a substitute for other drug or pain	162	64	8
Kratom Dosing	93	54	9
Kratom Regulation	297	79	4
Polydrug Use	85	29	10
Reports psychiatric symptoms or history	46	35	5
Quitting Kratom	64	33	3
Total Posts	4,333	1,389	215
Total Percentage for Each Topic			
Adverse Health Effects	2.12 %	4.54%	8.37%
Desirable Health Effects	1.64%	1.22%	0.93%
Kratom Addiction	3.07%	3.67%	3.72%
Kratom as a substitute for other drug or pain	3.74%	4.61%	3.72%
Kratom Dosing	2.15%	3.89%	4.19%
Kratom Regulation	6.85%	5.69%	1.86%
Polydrug Use	1.96%	2.09%	4.65%
Reports psychiatric symptoms or history	1.06%	2.52%	2.33%
Quitting Kratom	1.48%	2.38%	1.40%

tory. Conversely, the proportion of posts about kratom regulation and desirable health effects have decreased over time. Discussions about kratom addiction and using kratom as a substitute for other drugs or pain have remained relatively stable on subreddits that fall in the general discussion category. These findings are consistent with other papers on the topic and highlight that these new kratom drinks have not fundamentally changed kratom users' experiences (Smith et al., 2021; Garcia-Romeu et al., 2020; Coe et al., 2019). Rather, the results in this section, and Figure 1 in particular, highlight that these new kratom drinks have increased the amount of interest in kratom over time.

4 Conclusion

This study examines recent trends in kratom use in the U.S. using Reddit data from 2017 to 2024. Our findings indicate a growing interest in kratom drinks, correlating with an overall increase in discussions about kratom. Posts in subreddits focused on kratom addiction have also increased over time. Moreover, general discussion subreddits highlight an increasing number of top posts concerning adverse health effects, kratom dosing, polydrug use, and psychological factors related to kratom use.

This analysis was not intended as an in-depth qualitative study but rather aimed to utilize a relevant yet underused data source to provide additional insights into trends in kratom

use among adults on social media and gather detailed self-reported information. Our analysis highlights several key factors for the future monitoring and regulation of kratom. Most importantly, new kratom drinks should be clearly labeled to indicate their addictive potential. A significant issue during the opioid crisis was the lack of awareness among users and prescribers about the addictive nature of opioids. Similarly, many individuals consuming kratom drinks are unaware of their addictive potential. Our analysis may also help in developing or refining new surveys or qualitative interview questions related to kratom use for future research.

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5 Appendix

Table A1: Subreddit Information

Subreddit Name	Date	Type of Subreddit	Subscribers	Reddit URL
kratom	2009-04-21	General Discussion	169737	/r/kratom/
Kava	2012-01-19	Complement/Substitute	36596	/r/Kava/
quittingkratom	2013-07-11	Kratom Addiction	43031	/r/quittingkratom/
Kanna	2014-05-06	Complement/Substitutes	13605	/r/Kanna/
KratomKorner	2014-07-26	General Discussion	35940	/r/KratomKorner/
Kratom_Info_Exchange	2016-08-15	General Discussion	11262	/r/Kratom_Info_Exchange/
KratomGarden	2017-01-01	Growing/Buying Kratom	10705	/r/KratomGarden/
Kratomtree	2017-01-23	Growing/Buying Kratom	2467	/r/Kratomtree/
recreationalkratom	2017-04-17	General Discussion	10459	/r/RecreationalKratom/
kratomm	2017-09-28	General Discussion	11756	/r/Kratomm/
kratomeu	2018-06-09	General Discussion	1575	/r/kratomeu/
KhatGrowing	2020-02-02	Complement/Substitutes	1809	/r/KhatGrowing/
KratomDeals	2020-07-11	Growing/Buying Kratom	2721	/r/KratomDeals/
kratomreports	2021-05-08	Growing/Buying Kratom	3406	/r/kratomreports/
magicplantsexchanges	2021-09-12	Complement/Substitutes	8138	/r/magicplantsexchanges/
MitraMikeExtractz	2022-02-12	Growing/Buying Kratom	1701	/r/MitraMikeExtractz/
Vendorsofkratom2	2022-07-07	Growing/Buying Kratom	5440	/r/Vendorsofkratom2/
Kratom_Extractors	2022-09-23	Growing/Buyuing Kratom	946	/r/Kratom_Extractors/
kratomreview100	2022-12-04	Growing/Buying Kratom	4223	/r/kratomreview100/
Quittingfeelfree	2023-01-13	Kratom Addiction	1948	/r/Quittingfeelfree/